JOSH MONTIERTH

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SUMMARY

Engineer and graduate researcher specializing in digital signal processing and wireless communication systems. Experienced in designing, implementing, and evaluating wireless systems, with expertise spanning digital communications, channel modeling, algorithm design, and electromagnetic propagation. Brings both hands-on engineering experience and graduate research to practical, technically demanding problems.

EDUCATION

Brigham Young University *Expected April 2027 | Provo, UT*

Master of Science in Electrical and Computer Engineering, 4.0 GPA

- 2026 recipient of the National Science Foundation Graduate Fellowship to further research in wireless innovation
- Thesis work focused on wireless communication and embedded RF systems

Brigham Young University *April 2026 | Provo, UT*

Bachelor of Science in Electrical Engineering, Summa Cum Laude, 4.0 GPA

- Minors in Computer Science and Mathematics

PROFESSIONAL EXPERIENCE

Network Research Scientist, Riverside Research *September 2026–Present | Remote*

- Team lead and designer for wireless defense technology development
- Design and implement wireless algorithms and protocols to enable new functionality in existing commercial hardware

Research Assistant, BYU Network Enhanced Technologies Lab *November 2023–Present | Provo, UT*

- Authored three peer-reviewed publications on novel wireless protocols and presented them at international conferences
- Lead researcher developing new wireless systems, algorithms, and protocols on SDRs and embedded devices
- Collaborated on multi-university research on communications for underwater autonomous vehicles

Lead RF Engineer, Rincon Research Corporation *September 2025–April 2026 | Provo, UT*

- Lead engineer for BYU capstone project (250+ hours of product development) sponsored by Rincon Research Corporation
- Designed and implemented a low-cost RF sensing system for terrestrial detection of spaceborne imaging radar

Course Instructor, BYU College of Engineering *April 2025–August 2025 | Provo, UT*

- Instructed sophomore computer systems course with topics on computer architecture and C language fundamentals
- Developed curriculum for new university circuits course including 12 hands-on labs and taught the pilot class

Teaching Assistant, BYU Physics Department *January 2024–April 2025 | Provo, UT*

- Taught physics course on electromagnetics to classes of 100+ undergraduate students

TECHNICAL SKILLS

- **RF & DSP:** Software-defined radio (USRP, ADALM-Pluto, GNU Radio), digital communications, waveform and protocol design, channel modeling, RF systems, Wi-Fi, LoRa, UWB, algorithm design, error control codes
- **Programming:** C, C++, Python (including PyTorch), MATLAB (including Simulink), Java, JavaScript
- **Design & Fabrication:** Autodesk Inventor (Certified Professional), Fusion 360, 3D printing, CNC machining, laser cutting
- **Languages:** Portuguese (fluent), Spanish (working proficiency)

RELEVANT UNIVERSITY COURSEWORK

• Electromagnetic Propagation • Digital Signal Processing • Transmission Line Theory • Embedded Programming
• Machine Learning • Quantum Optics • Digital Communication Theory • Stochastic Processes • Error Control Coding

ACTIVITIES & LEADERSHIP

- Eagle Scout, Tau Beta Pi Member
- 2024–2026: Volunteer public speaker and organizer of community events and service activities for 1000+ individuals
- 2021–2023: Led 100 volunteers in southern Brazil, developed KPIs, and improved team alignment and productivity
- 2016–2020: Funded college education through personal landscaping business

PERSONAL INTERESTS

• Competitive classical piano (12 years) • Science fiction writing •3D design and fabrication